

Implement Greedy Search Algorithm for Selection Sort.

Code :-

```
arr = [64, 25, 12, 22, 11]
```

```
n = len(arr)
```

```
for i in range(n):
```

```
    pos = i
```

```
        for j in range(i+1, n):
```

```
            if arr[j] < arr[pos]:
```

```
                pos = j
```

```
arr[i], arr[pos] = arr[pos], arr[i]
```

```
print("Sorted array:", arr)
```

Theory:- ☒ Aim

To sort a list of elements using Selection Sort based on Greedy approach.

☒ Theory

Selection Sort is a simple sorting technique based on the greedy method.

In this algorithm, at each step, the smallest element is selected from the unsorted part of the list and placed at its correct position.

It divides the array into two parts:

1. Sorted part
2. Unsorted part

Initially, the sorted part is empty. The algorithm repeatedly selects the minimum element from the unsorted part and swaps it with the first unsorted element.

Since it always chooses the best (minimum) element at each step, it follows the greedy strategy.

☒ Algorithm Steps

1. Start
2. Take an array of size n
3. Repeat from $i = 0$ to $n-1$
 - Assume current index i as position of smallest element
 - For $j = i+1$ to $n-1$
 - If $arr[j] < arr[pos]$
→ update position $pos = j$
 - Swap $arr[i]$ with $arr[pos]$
4. Repeat until the array is sorted
5. Stop